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Module - VII : Reliability

If a component is put into operation at some specified time, say $t=0$, and if T is the time until it fails or ceases to function properly, T is called the life length or time to failure of the component.

Obviously $T(\geq 0)$ is a continuous random variable with some probability density function $f(t)$.

Then the Reliability (or) Reliability function of the component at time t denoted by $R(t)$ is defined as $R(t) = P(T > t)$ or $1 - P(T \leq t)$
 $F(T)$ - Cumulative distribution = $1 - R(t)$. Sum of T &

$$F(t) = \int_0^t f(t) dt \quad (\text{or}) \quad R(t) = 1 - \int_0^t f(t) dt = \int_t^{\infty} f(t) dt$$

$\therefore F(0) = 0$ & $F(\infty) = 1$ by the property of cdf,
 $R(0) = 1$ & $R(\infty) = 0$, i.e. $0 \leq R \leq 1$.

Also $\frac{d}{dt} F(t) = f(t)$ (or) $f(t) = - \frac{d}{dt} R(t)$

Conditional probability of failure in the interval $(t, t + \Delta t)$, given that the component has survived upto time t , $P\{t \leq T \leq t + \Delta t \mid T \geq t\} = \frac{f(t) \Delta t}{1 - F(t)}$
 $= \frac{f(t) \Delta t}{R(t)}$.

The conditional probability of failure per unit (2) time is given by $\frac{f(t)}{R(t)}$ and is called the instantaneous failure rate or Hazard Fun of the component, denoted by $\lambda(t)$. (a) $\lambda(t) = \frac{f(t)}{R(t)}$
 [After some modifications, $f(t) = \lambda(t) e^{-\int_0^t \lambda(t) dt}$]

Expected $\lambda(t) = - \frac{R'(t)}{R(t)}$

$$\int_0^t \frac{R'(t)}{R(t)} dt = - \int_0^t \lambda(t) dt$$

$$\log(R(t)) \Big|_0^t = - \int_0^t \lambda(t) dt$$

$$\log_e R(t) = - \int_0^t \lambda(t) dt \quad (b) \quad R(t) = e^{-\int_0^t \lambda(t) dt}$$

$$\therefore f(t) = \lambda(t) \cdot e^{-\int_0^t \lambda(t) dt}$$

The expected value of the time to failure T , denoted by $E(T)$ and variance of T denoted by σ_T^2 are two important parameters frequently used to characterise reliability.

$E(T)$ is called Mean Time to Failure (MTF).

$$MTF = \int_0^{\infty} R(t) dt, \quad \text{Var}(T) = \int_0^{\infty} t^2 f(t) dt - (MTF)^2$$

Conditional reliability is another concept used to describe the reliability of a component or system following a wear in period (burn-in period) or after a warranty period.

$$R(t/T_0) = P\{T > T_0 + t / T > T_0\}$$

$$= e^{-\int_{T_0}^{T_0+t} \lambda(t) dt}$$

1. The density fun of the time to failure in years of ③ the gizmos [for use on widgets] manufactured by a certain company is given by $f(t) = \frac{200}{(t+10)^3}$, $t \geq 0$. (a) Derive the Reliability fun and determine the ^{MTTF} reliability for the first year of operation. (b) Compute ^{MTTF} MTF, (c) what is the design life for a reliability 0.95? (d) Will a one year burn-in-period improve the reliability in part (a)? If so what is the new reliability.

$$(a) f(t) = \frac{200}{(t+10)^3}, \quad t \geq 0 \quad R(t) = \int_t^{\infty} f(t) dt$$

$$R(t) = \left[200 \times \frac{(t+10)^{-2}}{-2} \right]_t^{\infty} = -100 \left[\frac{1}{(t+10)^2} \right]_t^{\infty} = \frac{100}{(t+10)^2}$$

$$R(1) = \frac{100}{(1+10)^2} = 0.8264$$

$$(b) MTTF := \int_0^{\infty} R(t) dt = \int_0^{\infty} \frac{100}{(t+10)^2} dt = \left[100 \frac{(t+10)^{-1}}{-1} \right]_0^{\infty} \\ = \frac{100}{10} = 10 \text{ years}$$

(c) Design life is the time to failure (t_D) that corresponds to a specified reliability. Now it is required to find t_D corresponding to $R = 0.95$.

$$\therefore \frac{100}{(t_D + 10)^2} = 0.95 \quad (\text{w } (t_D + 10)^2 = 100.2632)$$

$$\therefore t_D = 0.2598 \text{ year (or) } 95 \text{ days.}$$

$$(d) R(t/1) = \frac{R(t+1)}{R(1)} = \frac{100/(t+1+10)^2}{100/(1+10)^2} = \frac{11^2}{(t+1)^2}$$

$$\text{Now, } R(t/1) > R(t) \text{ is } \frac{121}{(t+1)^2} > \frac{100}{(t+10)^2}$$

$$(\text{w } \frac{(t+10)^2}{(t+1)^2} > \frac{100}{121} \quad (\text{w } \frac{t+10}{t+1} > \frac{10}{11} \quad (\text{w } 11t > 10t$$

One year burn-in-period will improve which is true as $t > 10$.

$$R(1/1) = \frac{121}{(1+1)^2} = 0.8403 > 0.8264 \text{ the reliability}$$

2) The time to failure in operating hours of a critical solid state power unit has the hazard rate $\lambda(t)$ (4)

$\lambda(t) = 0.003 \left(\frac{t}{500} \right)^{0.5}$ for $t \geq 0$. (a) What is the reliability if the power unit must operate continuously for 50 hours? (b) Determine the design life if a reliability of 0.9 is desired. (c) Compute the MTTF.

(d) Given that the unit has operated for 50 hours, what is the probability that it will survive a second 50 hours of operation?

Soln: $-\int_0^t \lambda(u) du$

$$(a) R(50) = e^{-\int_0^{50} 0.003 \left(\frac{t}{500} \right)^{0.5} dt}$$

$$= e^{-\frac{0.003}{\sqrt{500}} \cdot \frac{2}{3} [t^{3/2}]_0^{50}} = e^{-0.03162} = 0.9689$$

$$(b) R(t_D) = 0.90 \quad (c) e^{-\int_0^{t_D} 0.003 \sqrt{\frac{t}{500}} dt} = 0.90$$

$$-\int_0^{t_D} \frac{0.003}{\sqrt{500}} t^{1/2} dt = -0.10536$$

$$\frac{0.003}{\sqrt{500}} \cdot \frac{2}{3} t_D^{3/2} = 0.10536 \quad (d) t_D = 111.54 \text{ H.}$$

$$(c) \text{MTTF} = \int_0^{\infty} R(t) dt = \int_0^{\infty} e^{-\left(\frac{0.003}{\sqrt{500}} \times \frac{2}{3} \times t^{3/2}\right)} dt$$

$$= \int_0^{\infty} e^{-at^{3/2}} dt \quad \text{where } a = \frac{0.003 \times 2}{3 \times \sqrt{500}}$$

$$= \int_0^{\infty} e^{-x} \cdot \frac{2}{3a^{2/3}} x^{-1/3} dx \quad \text{on putting } x = at^{3/2}$$

$$= \frac{2}{3a^{2/3}} \Gamma\left(\frac{2}{3}\right) = \frac{2}{3a^{2/3}} \times \frac{3}{2} \Gamma\left(\frac{5}{3}\right) = 451.65 \text{ h.}$$

$$(d) P\{T \geq 100 / T \geq 50\} = \frac{P(T \geq 100)}{P(T \geq 50)} = \frac{R(100)}{R(50)}$$

$$= \frac{e^{-\int_{50}^{100} R(t) dt}}{e^{-\int_{0}^{50} R(t) dt}} = e^{-\left[\frac{0.002}{\sqrt{500}} 100^{3/2} - \frac{0.002}{\sqrt{500}} 50^{3/2}\right]}$$

$$= e^{-(0.08944 - 0.03162)} = 0.9438$$

Some Special Failure Distributions:

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(1) The exponential distribution:

$$\text{pdf: } f(t) = \lambda e^{-\lambda t}, \quad R(t) = e^{-\lambda t}, \quad \lambda(t) = \lambda$$

$$\text{MTTF} = E(T) = \frac{1}{\lambda}; \quad \text{Var}(T) = \frac{1}{\lambda^2}$$

$$R(t/T_0) = \frac{e^{-\lambda(T_0+t)}}{e^{-\lambda T_0}} = e^{-\lambda t}$$

(2) The Weibull Distribution:

$$f(t) = \alpha \beta t^{\beta-1} e^{-\alpha t^\beta}, \quad t \geq 0 \quad (\text{or})$$

$$f(t) = \frac{\beta}{\theta} \left(\frac{t}{\theta} \right)^{\beta-1} e^{-\left(\frac{t}{\theta} \right)^\beta}, \quad \theta > 0, \beta > 0, t \geq 0.$$

θ - characteristic life (or) scale parameter, $\alpha = \frac{1}{\theta^\beta} \cdot \beta$ is called the shape parameter.

$$R(t) = e^{-x} = e^{-\left(\frac{t}{\theta} \right)^\beta}, \quad \lambda(t) = \frac{\beta}{\theta} \left(\frac{t}{\theta} \right)^{\beta-1}$$

$$\text{MTTF} = \theta \Gamma\left(1 + \frac{1}{\beta}\right), \quad \text{Var}(T) = \theta^2 \left\{ \Gamma\left(1 + \frac{2}{\beta}\right) - \left[\Gamma\left(1 + \frac{1}{\beta}\right) \right]^2 \right\}$$

$$R(t/T_0) = \frac{\left[-\left(\frac{t+T_0}{\theta} \right)^\beta + \left(\frac{T_0}{\theta} \right)^\beta \right]}{e^{-\left(\frac{T_0}{\theta} \right)^\beta}}$$

(3) Normal Distribution:

$$-\frac{(t-\mu)^2}{2\sigma^2}, \quad -\infty < t < \infty$$

$$f(t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$$

$$MTTF = \mu \quad \text{and} \quad \text{Var}(\tau) = \sigma^2$$

A manufacturer determines that, on the average, a television set is used 1.8 hours per day. A one-year warranty is offered on the picture tube having a MTTF of 2000 hours. If the distribution is exponential, what percentage of the tubes will fail during the warranty period?

$R(t) = e^{-\lambda t}$ where λ is the failure rate.

$$MTTF = 2000 \text{ H} \quad \int_0^{\infty} e^{-\lambda t} dt = 2000 \quad (1) \quad \frac{1}{\lambda} = 2000$$

$$\lambda = 0.0005 \text{ hours}^{-1}$$

$$P(T \leq 1) = P(T \leq 365 \times 1.8) \quad [\because \text{The TV is operated } 1.8 \text{ H per day}]$$

$$= 1 - P\{P > 657\} = 1 - e^{-0.0005 \times 657} = 0.28$$

$\therefore$ 28% of the tubes will fail during the warranty period.

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Reliability of Systems:

Serial Configuration: Series or nonredundant Configuration is one in which the components of the system are connected in series (or serially). $C_1 \rightarrow C_2 \rightarrow \dots \rightarrow C_n$

All components must run for the system to function.

$C_1, C_2, \dots, C_n$ — n independent components.

$R_1, R_2, \dots, R_n$ — Reliability of the components.

$$R_S(t) = R_1(t) \times R_2(t) \times \dots \times R_n(t) \leq \min \{R_1(t), R_2(t), \dots, R_n(t)\}$$

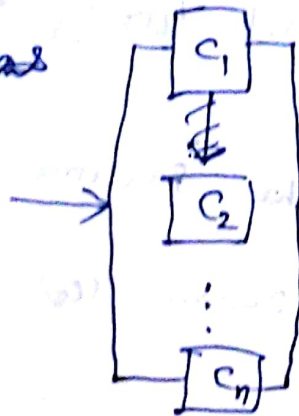
Note: $0 < R_i < 1$.

① If each component has a constant failure rate λ , then $R_S(t) = e^{-\lambda_1 t} e^{-\lambda_2 t} \dots e^{-\lambda_n t} = e^{-\lambda_S t}$; $\lambda_S = \sum_{i=1}^n \lambda_i$.

② If the components follow the Weibull distribution failure law parameters β_i & θ_i , then $R_S(t) = e^{-\sum_{i=1}^n \left(\frac{t}{\theta_i}\right)^{\beta_i}}$.

Parallel Configuration:

Parallel or redundant configuration is one in which the components of the system are connected in parallel. as



Note: In Parallel Configuration, all components must fail for the system to fail.

$$R_p = 1 - (1 - R_1)(1 - R_2) \dots (1 - R_n) \\ \geq \max \{R_1, R_2, \dots, R_n\}$$

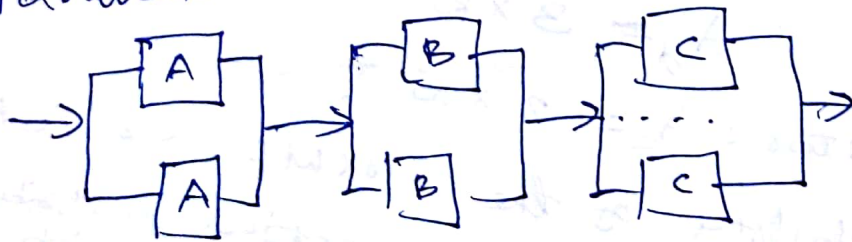
For Two Component system, in parallel having $\oplus$ constant failure rate, $R_i(t) = 1 - e^{-\lambda_i t}$; $i=1,2$.

$$R_p(t) = 1 - (1 - e^{-\lambda_1 t})(1 - e^{-\lambda_2 t})$$

$$= e^{-\lambda_1 t} + e^{-\lambda_2 t} - e^{-(\lambda_1 + \lambda_2)t}$$

$$MTTF = \int_0^{\infty} R_p(t) dt = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} - \frac{1}{\lambda_1 + \lambda_2}$$

Parallel Series Configuration (or) Low-level-redundancy 1:



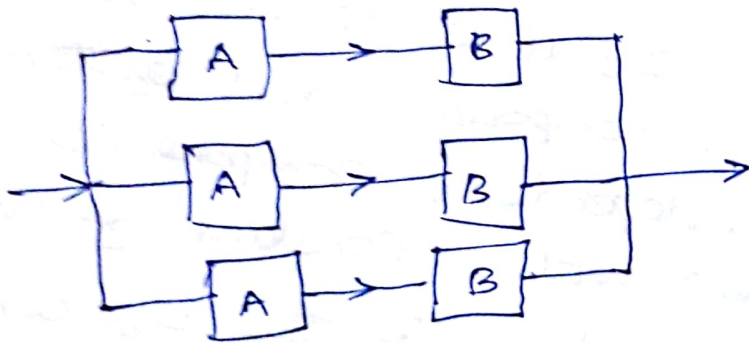
$n=2$ in the diagram
 $m=3$

Reliability of each of the subsystems $= 1 - (1 - R)^n$

Reliability of the low-level redundancy is given by

$$R_{Low} = \{1 - (1 - R)^n\}^m$$

Series Parallel Configuration:



A system in which m subsystems are connected in parallel where each subsystem has n components connected in series.

Reliability of each component (each subsystem $m=2, n=3$).
 $= R^n$

System reliability for the high level redundancy

$$R_{\text{high}} = 1 - (1 - R^n)^m$$

Note: When $m = n = 2$; $R_{\text{low}} > R_{\text{high}}$

$$R_{\text{low}} - R_{\text{high}} = 2R^2 \cdot (1 - R)^2 \geq 0.$$

① An electronic circuit consists of 5 silicon transistors, 3 silicon diodes, 10 composition resistors and 2 ceramic capacitors connected in series configuration. The hourly failure rate of each component is given below.

Silicon transistor: $\lambda_t = 4 \times 10^{-5}$

Composition resistor: $\lambda_r = 2 \times 10^{-7}$

Silicon diode: $\lambda_d = 3 \times 10^{-5}$

Ceramic capacitor: $\lambda_c = 2 \times 10^{-7}$

Calculate the reliability of the circuit for 10H, when the components follow exponential distribution.

$$R_s(t) = e^{-(5\lambda_t + 3\lambda_d + 10\lambda_r + 2\lambda_c)t}$$

$$R_s(10) = e^{-(20 + 9 + 200 + 40) \times 10^{-7}} = e^{-0.0269}$$

$$= 0.9735$$

② Two parallel, identical & independent components have constant failure rate. If it is desired that $R_S(1000) = 0.95$, find the component and system MTTF. In addition to the independent components is a common mode component with a constant failure rate of 0.00001 is connected to the system, find the new system MTTF.

Soln: Let R and λ be the reliability and failure rate of each of the two independent

components. Then $R(t) = e^{-\lambda t}$

Since the two components are connected in parallel

$$R_S(t) = 1 - [1 - R(t)]^2 = 1 - \{1 - e^{-\lambda t}\}^2 = 2e^{-\lambda t} - e^{-2\lambda t}$$

$$2 \cdot e^{-1000\lambda} - e^{-2000\lambda} = R_S(1000) = 0.95$$

put $x = e^{1000\lambda}$ (or) $x^2 - 2x + 0.95 = 0$ (9)

(or) $x = \frac{2 \pm \sqrt{4 - 3.8}}{2} = 1.22361$ (or) 0.77639
 $e^{-1000\lambda} =$

$-1000\lambda = 0.20181$ (or) -0.25310

$\lambda = 0.0002531$ (Reject the -ve value of λ)

Component MTTF = $\frac{1}{\lambda} = \frac{1}{0.0002531} = 3951$.

and system MTTF = $\int_0^{\infty} R_S(t) dt = \int_0^{\infty} (2e^{-\lambda t} - e^{-2\lambda t}) dt$
 $= \frac{3}{2\lambda} = 5926.5$

Let λ be the failure rate of the common mode component connected in series with the already existing redundant system.

Then the system reliability becomes

$$R'_S(t) = R_S(t) \cdot R'(t) \quad \text{where } R'(t) \text{ is the}$$

Reliability of the common mode component.

$$R'_S(t) = (2e^{-\lambda t} - e^{-2\lambda t}) e^{-\lambda' t}$$

$$\therefore R'_S(1000) = (2e^{-0.2531} - e^{-0.5062}) e^{-0.01}$$

$$(\because \lambda = 0.0002531, \lambda' = 0.00001)$$

$$\therefore R'_S(1000) = 0.94$$

Now the new system MTTF is given by

$$(\text{MTTF})' = \int_0^{\infty} [2e^{-(\lambda+\lambda')t} - e^{-(2\lambda+\lambda')t}] dt$$

$$= \frac{2}{\lambda+\lambda'} - \frac{1}{2\lambda+\lambda'} = \frac{2}{0.0002531} - \frac{1}{0.0005162}$$

$$= 5664.4$$

Maintainability :

Let $R(t)$ and $R_M(t)$ be the reliability of a system without & with maintenance.

$$R_M(t) = R(t) \quad \text{for } 0 \leq t < T$$
$$= R(T) \quad \text{for } t = T$$

After performing the maintenance operation at t , the system becomes good as new.

Hence $R_M(t) = R(T) \cdot R(t-T)$ for $T \leq t < 2T, \dots$

In general $R_M(t) = [R(T)]^n \cdot R(t-nT)$ for $nT \leq t < (n+1)T$

$$MTTF = \frac{\int_0^\infty R(t) dt}{1 - R(T)}$$

If the system has a constant failure rate λ , then

$$R(t) = e^{-\lambda t} \quad \text{and} \quad R_M(t) = (e^{-\lambda T})^n \cdot e^{-\lambda(t-nT)} \quad \text{for}$$
$$nT \leq t < (n+1)T$$
$$= e^{-\lambda t}$$

(W) In case of constant failure rate, preventive maintenance does not improve the reliability of the system.

⊗ If the failure distribution of the system is Weibull for non-maintained system,

$$R(t) = e^{-\left(\frac{t}{\theta}\right)^{\beta}}$$

$$R_M(t) = e^{-n\left(\frac{t}{\theta}\right)^{\beta}} e^{-n\left(\frac{t-nT}{\theta}\right)^{\beta}}$$

To examine the effect of preventive maintenance,

$$\frac{R_M(t)}{R(t)} = \frac{R_M(nT)}{R(nT)} \text{ we get } \beta > 1. \quad \Downarrow \quad \therefore \text{Will improve the reliability of the system}$$

Repair Maintenance - Maintainability :

$$\text{Time to repair: } M(t) = \int_0^t m(t) dt$$

$$\text{MTTR: } \int_0^t t \times m(t) dt$$

$\mu(t)$ - Instantaneous repair rate.

(4)

$$\mu(t) = \frac{P\{t \leq T \leq t + \Delta t\}}{P\{T > t\}} = \frac{m(t)}{1 - M(t)}$$

$$\therefore m(t) = \mu(t) \cdot e^{-\int_0^t \mu(t) dt}$$

(*) If $\mu(t) = \mu$ (Constant) then $m(t) = \mu e^{-\mu t}$, so

$$M(t) = 1 - e^{-\mu t}$$

$$MTTR: \int_0^{\infty} t m(t) dt = \frac{1}{\mu}$$

Availability:

If $A(t)$ is the point availability of a component

$$\text{or system, } A(t_2 - t_1) = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} A(t) dt.$$

is called the interval availability (or) mission availability.

In particular, in $(0, T)$.

$$A(t) = \frac{1}{T} \int_0^T A(t) dt.$$

Note: $\lim_{T \rightarrow \infty} A(t)$ is called steady state (or) long run availability A (or) $A(\infty)$.

Component unavailability: $\bar{A}(\infty) = \frac{\lambda}{\lambda + \mu}$ λ - component constant failure rate

* If Failure of the component follows Poisson distribution $P\{N(t) = r\} = \frac{e^{-\lambda t} \cdot (\lambda t)^r}{r!}$, $r = 0, 1, 2, \dots$

Mean time between failure $\cdot \frac{1}{\lambda}$.

Note: $A(\infty) = 1 - \frac{\lambda}{\mu}$

System Availability:

$$A_s(t) = 1 - \{1 - A_1(t)\} \{1 - A_2(t)\} \dots \{1 - A_n(t)\}$$

If a device has a failure rate of $\lambda(t) = \frac{(0.015 + 0.02t)}{2}$ years³
 where t is in years.

- (a) Calculate the reliability for a 5 year design life, assuming that no maintenance is performed.
 (b) Calculate the reliability for 5 years assuming that annual preventive maintenance restores the device to an as-good-as-new condition.
 (c) Repeat part (b) assuming that there is a 5% chance that the preventive maintenance will cause immediate failure.

Soln: (a). $R(t) = e^{-\int_0^t \lambda(t) dt}$ $\therefore R(5) = e^{-\int_0^5 (0.015 + 0.02t) dt}$
 $= e^{-0.325} = 0.7225$

- (b) Since annual preventive maintenance is performed, there will be 4 preventive maintenances in the first 5 years.

$$R_M(t) = \{R(T)\}^n \times R(t - nT) \text{ after } n \text{ maintenances.}$$

Here $t=5$, $T=1$, & $n=4$.

$$R_M(5) = \{R(1)\}^4 \cdot R(5-4) = \{R(1)\}^5 = \left[e^{-\int_0^1 (0.015 + 0.02t) dt} \right]^5 = [e^{-0.025}]^5 = 0.8825$$

(c) n.c.

(c) $P \{ \text{preventive maintenance causes immediate failure} \} = 0.05$

$\therefore P \{ \text{The device survives after each preventive maintenance} \}$
 $= 0.95$

As there are 4 maintenances

$R_M(t) = P_M(t) \text{ without breakdown} \times \text{probability of no breakdown in } t \text{ years.}$

$$= 0.8825 \times (0.95)^4 = 0.7182$$

(2) The time to failure (in hours) of a piece of equipment is uniformly distributed over $(0, 1000)$ hours.

(a) Determine MTTF. (b) Determine the MTTF if preventive maintenance will restore the system to as good as new condition and is performed every 100 operating hours.

(c) Compare the reliability with & without preventive maintenance at 225 operating hours. Assume the 100 hour maintenance interval and a maintenance-induced failure probability of

0.0) each time preventive maintenance is performed. (12)

(d) Is there a significant improvement in reliability if a 50 hour preventive maintenance interval is assumed?

Soln: The pdf of the time to failure is given by

$$f(t) = \frac{1}{1000} \quad \text{in } 0 \leq t \leq 1000.$$

$$(a) \text{MTTF} = \int_0^{1000} \frac{t}{1000} dt = 500 \text{ Hours.}$$

$$(b) (\text{MTTF})_M = \frac{\int_0^{1000} R(t) dt}{\int_0^{1000} [1 - R(1000)]} \quad \text{where } R(t) = \int_0^t \frac{1}{1000} dt = 1 - 0.0001t$$

$$= \frac{\int_0^{1000} [1 - 0.0001t] dt}{1 - (1 - 0.0001 \times 1000)} = \frac{1}{0.1} \left\{ t - 0.0001 \frac{t^2}{2} \right\}_0^{1000} = 950 \text{ H.}$$

$$(c) R(225) = 1 - 0.0001 \times 225 = 0.775$$

$$R_M(225) = \{R(T)\}^2 \times R(t - 2T) \times (1 - p)^2$$

$n=2$, preventive maintenance in (0, 225).

$$R_M(225) = \{R(1000)\}^2 \times R(25) \times (1 - 0.01)^2$$

$$= 0.774 \quad \& \quad R(225) \geq R_M(225)$$

6) If $T=50$, $n =$ the no of preventive maintenance $= 4$,

$$\therefore R'_n(25) = \{R(50)\}^4 \times R(225-200) = 0.794$$

If there is no maintenance induced failure probability the reliability is improved when $T=50$.

3) A new computer has a constant failure rate of 0.02 per day (assuming continuous use) and a constant repair rate of 0.1 per day.

- (a) Compute the interval availability for the first 30 days and the steady state availability.
- (b) Determine the steady state availability if a standby unit is purchased. Assume no failures in standby.
- (c) If both units are active, what is the steady state availability?

slms $\lambda = 0.02 / \text{day}$; $\mu = 0.1 / \text{day}$.



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$$(a) A_1(T) = \frac{\mu}{\lambda + \mu} + \frac{\lambda}{(\lambda + \mu)^2} \times T \left\{ 1 - e^{-(\lambda + \mu)T} \right\}$$

$$A_1(30) = \frac{0.1}{0.12} + \frac{0.02}{(0.12)^2} \times 30 \left\{ 1 - e^{-0.12 \times 30} \right\}$$
$$= 0.8784$$

$$A(\infty) = \frac{\mu}{\lambda + \mu} = 0.8333$$

(b) for the standby redundant system,

$$A_S(\infty) = \frac{\lambda \mu + \mu^2}{\mu^2 + \lambda \mu + \lambda^2} = 0.9677$$

for the active redundant system

$$A_S(\infty) = 1 - \left\{ 1 - A(\infty) \right\}^2 = 0.9722$$